BSFL Commercial Launch Plan

Lockyer Valley, Queensland · April 2026 — December 2026

STARTUP BUDGET	REVENUE TARGET	LAUNCH WINDOW	PRIMARY PRODUCT
$500	**$25,000**	**90 days**	**Live BSFL**
hard ceiling	by 31 Dec 2026	Apr – Jun 2026	pet & poultry market

– The fastest path to $25k is selling live and dried BSFL direct to the reptile, poultry, and backyard chicken market — not bulk protein meal or frass. Live BSFL sell for $25–60/kg retail in Australia; bulk wholesale to pet stores at $15–25/kg. You already have the rearing knowledge. The first 90 days are entirely about legal structure, waste supply, and first customers. No equipment purchases needed beyond packaging materials and a used dehydrator.

Business Structure

You are bringing in a partner, so a sole trader structure is not appropriate. Two options suit your priorities:

STRUCTURE	SETUP COST	KEY ADVANTAGE	KEY RISK
Partnership (registered ABN)	$0 (free ABN registration)	Simplest to set up. Both partners share income/loss on personal tax returns.	Joint and several liability — each partner personally liable for all business debts. Harder to license IP later.
Pty Ltd company (recommended)	$576 ASIC registration + ~$500–800 solicitor for constitution	Limited liability protects personal assets. Clean equity split. Far easier to license protocol and attract future partners or investors.	Higher setup cost. Annual ASIC review fee ($59). Requires a company constitution.

– Recommendation: Register a Pty Ltd company. The $576 ASIC fee plus a professionally reviewed shareholders agreement (~$500–800) is the single most important money you will spend. It determines what happens if one partner wants to exit, who owns the rearing protocol IP, how decisions are made, and how profit is split. Do not proceed on a handshake. If budget is the constraint, start as a partnership with a written partnership agreement drafted by a solicitor, then convert to Pty Ltd at the 6-month mark. Equity split: 51% (you) / 49% (partner) — see Partner Roles section for full rationale.

The 10-Step Launch Plan

PHASE 1 **Weeks 1–4 · Setup & Compliance** **$0 revenue · ~$185 spend**

1 **Register company or partnership ABN**

If forming a Pty Ltd: register at asic.gov.au ($576). If starting as a partnership: register a partnership ABN at abr.gov.au (free). Both take under 30 minutes online. Do this on day one — it unlocks invoicing, banking, and GST registration.

Free–$576 **Day 1** **abr.gov.au / asic.gov.au**

2 **Register a business name through ASIC ($42)**

Choose something short, local, and memorable. The Lockyer Valley provenance is a genuine marketing asset — names like 'Valley Grubs', 'Lockyer Larvae', or 'Gatton Bug Farm' communicate origin and trustworthiness. Check availability at asic.gov.au first. This name goes on every invoice, label, and market banner.

$42 / 1 year **Week 1**

3 **Draft and sign a partnership agreement or company shareholders agreement**

This is the most important document in the business. It must cover: profit split, decision-making authority, IP ownership (especially the rearing protocol), what happens if one partner exits, and how the business is dissolved. Budget $500–800 for a solicitor review. Do not operate without this.

$500–800 solicitor **Week 1–2** **Non-negotiable**

4 **Call Lockyer Valley Regional Council Environmental Health (1300 005 872)**

Describe your operation: food waste intake, insect rearing, live animal sales. Ask whether any permit or development approval is required on your existing property. In most rural and agricultural zones, small-scale insect farming is permitted use — but get this confirmed in writing via email follow-up. Free, 20 minutes.

Free **Week 1–2**

5 **Contact Biosecurity Queensland — DAF 13 25 23**

Ask one specific question: what food waste streams are approved for rearing BSFL destined for sale as poultry and pet feed in Queensland? Vegetable and fruit waste is unambiguously permitted. Get confirmation in writing. This protects you if a customer or supplier later questions your substrate sources.

Free **Week 2**

6 **Open a dedicated business bank account**

ING, Up, or Commonwealth Bank Business accounts are all free with no monthly fee. Required for clean bookkeeping and EFT payments from wholesale customers. Bring your ABN or company registration. Set up Wave Accounting (free) at the same time — GST registration from day one lets you claim back tax on equipment purchases.

Free **Week 1–2**

7 **Secure at least one food waste supply agreement**

Target Gatton, Laidley, or Toowoomba cafes, bakeries, and fruit/vegetable wholesalers. Lead with the environmental pitch: free collection, landfill diversion, zero cost to them. One medium café generates 50–150 kg of vegetable scraps per week. Get a written agreement (email is sufficient): waste type, frequency, your collection responsibility. Aim for tipping fee income ($30–80/tonne) once you have operating history.

Free feedstock **Week 2–4**

PHASE 2 Weeks 5–8 · First Products & First Sales

Target $2,000–3,500 · ~$215 spend

8 **Buy a USB microscope (~$30) and document your operation from day one**

A 50–200x USB microscope (Amazon, ~$25–40) is the highest-ROI purchase in the budget. Photograph larvae at magnification and post images in reptile and poultry Facebook groups — almost no small Australian producer does this, and it positions you as expert rather than just another supplier. It justifies premium pricing, builds trust for standing orders, and doubles as quality control. Separately, start a simple batch tracking spreadsheet: substrate source, harvest date, rearing temperature, yield weight. This costs nothing and simultaneously builds the licensed protocol documentation for your long-term plan.

~$30 **Week 5** **Highest ROI purchase**

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9 **Scale rearing to commercial volume using existing setup**

Using your existing rearing knowledge and infrastructure, target 5–10 kg of harvestable live larvae per week by week 8. No new equipment required — repurposed food-grade containers and a thermostatically controlled space. The Lockyer Valley averages 27–30°C in summer, eliminating heating costs during launch. Note: above 38°C, mortality accelerates sharply. Install a $20 thermometer and ensure shade and ventilation before November — a single hot week in December can wipe a production cycle at the worst possible time.

No new capital **Weeks 5–8** **Heat risk: plan for Nov–Mar**

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10 **Buy a used dehydrator and create packaging**

Source a second-hand food dehydrator on Facebook Marketplace or Gumtree for $50–70. Dried BSFL keep for 12+ months, eliminate live-delivery logistics, and sell at $70–100/kg — roughly 3x the live wholesale rate. Simultaneously, buy kraft zip-lock bags (500g and 1kg sizes, ~$40 for 100) and design labels in Canva (free): business name, ABN, product name, weight, substrate note, and QR code to your website. Build your website on GitHub Pages (free) with a custom domain (~$15/year via Namecheap) rather than Shopify — 'DM to order, pay by PayID' is standard for Australian insect farms at this scale.

~$120 total **Week 5–6** **GitHub Pages = free forever**

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11 **Pre-sell before first harvest — Facebook groups**

Post in Lockyer Valley and SEQ Facebook groups NOW, before you have product: 'Local BSFL farm launching in 6 weeks — who wants early access?' Collect names and PayID deposits. This costs nothing, discovers real demand immediately, and compresses time to first revenue by 4–6 weeks. Target groups: 'Bearded Dragon Owners QLD', 'Backyard Chickens Australia', 'Brisbane Reptile Keepers'. Sell 500g live packs at $18–22 for local pickup. Backyard chicken keepers are your best first audience — they buy larger quantities, are less price-sensitive than reptile specialists, and are abundant in the Lockyer Valley's agricultural community.

Zero cost **Start now** **$18–22 / 500g pack**

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12 **Visit 3–5 local pet and produce stores for wholesale supply — in person**

Visit pet stores and rural supplies in Gatton, Ipswich, and Toowoomba. Bring a sample pack and a one-page product sheet (printed at home, ~$2). Offer trial supply at $12–15/500g wholesale (they retail at $22–28). Position yourself as local, fresher than freight-cooled interstate stock, available on short notice. Lock in at least one standing order before your first commercial harvest — this gives you a production target and income certainty. A single store taking 2 kg/week at $24/kg wholesale = $2,500/month.

~$10 samples & printing **Weeks 7–8** **$24–30/kg wholesale**

PHASE 3

Months 3–6 · Scale & Second Revenue Stream

Target $8,000–12,000 · self-funded

13 **Launch dried BSFL and frass products**

With your dehydrator running, add dried larvae to your product range at $70–100/kg. This removes the live-delivery problem, expands your sales radius nationally, and allows longer storage. Simultaneously, bag and sell frass as organic fertiliser at $12–18/kg retail (in branded 1 kg kraft bags) and $8–10/kg bulk to farms. Frass requires zero additional processing — dry slightly, bag, label. At approximately 1 kg frass produced per kg of larvae, this doubles your revenue from the same rearing effort. Use nicer kraft bags with a hand-stamped label for the garden retail market; keep plain bulk bags for farm orders.

Dehydrator already owned Dried: $70–100/kg Frass: $12–18/kg retail

14 **Approach the Lockyer Valley Growers Association and local vegetable farms**

The Lockyer Valley produces roughly a third of Australia's vegetables. Growers here are already investing in organic fertiliser programs and are acutely aware of input costs. A single introduction at a grower network meeting — framing yourself as a local food waste processor producing certified organic soil amendment — puts you in front of 20–50 potential frass customers at once. The pitch is not 'buy my bug poo' but: 'locally sourced organic fertiliser with full substrate traceability, at below your current NPK program cost.' Contract sizes are 10–20x larger than individual retail sales. A 6-month frass supply contract with one market garden (500 kg/month at $10/kg) = $5,000.

Free — network meeting **Month 3–5** **$5,000 per 6-month contract**

15 **Sell at Lockyer Valley and Toowoomba farmers markets (fortnightly)**

Market stall fees are typically $30–60/day. Sell live larvae, dried larvae, and frass side by side. One well-positioned stall targeting $400–600/day generates $1,600–2,400/month. Market customers become your most loyal repeat buyers and most effective word-of-mouth source. The frass story alone draws the organic gardening crowd. The microscope photographs make exceptional point-of-sale signage — print an A3 enlargement of your best larval image and use it as your stall backdrop.

$30–60/day stall fee **Target $500/day** **Month 3–6**

16 **List on Gumtree, Facebook Marketplace, and approach online pet retailers**

Free listings on Gumtree and Facebook Marketplace generate consistent local inquiries at zero cost. For national reach, approach Australian online pet retailers (PetCircle, VetShopAustralia, specialist reptile stores) about wholesale supply agreements. A single mid-size online retailer taking 5 kg/week at $20/kg = $5,200/month. Lead times for supplier onboarding are typically 2–6 weeks — start these conversations in month 3 to have supply agreements live by month 5.

Free listings $20/kg wholesale online **Month 3–5**

17 **Begin chitin extraction from frass — sell to your existing grower customers**

Your frass already contains chitin from shed larval exoskeletons and pupal casings — approximately 5–8% of dried frass by weight. From month 4–5, run small-batch chitin extraction alongside your normal frass processing: a three-step chemical process using food-grade hydrochloric acid (demineralisation), sodium hydroxide (deproteinisation), and optional deacetylation to produce chitosan. All reagents are available from agricultural chemical suppliers. The extraction equipment needed at small scale is minimal — a stainless steel pot, pH paper, a food-grade filter cloth, and an oven. Total setup cost under $200. The first product is agricultural-grade chitin powder ($15–40/kg) sold directly to the vegetable growers who are already buying your frass. The pitch: 'This is the active ingredient in your frass that suppresses soil pathogens and triggers plant immune response — now concentrated.' For every 10 kg of dried frass processed, you recover approximately 0.6–0.8 kg of crude chitin powder. As volume increases and quality documentation builds, the same material — BSF-derived chitosan — commands $50–150/kg from specialty agricultural and nutraceutical buyers, and is described by international suppliers as the gold standard in chitosan quality due to its high deacetylation degree and consistent insect-origin substrate.

~$200 setup **Month 4–5 start** **$15–150/kg depending on grade** **Zero additional feedstock cost**

PHASE 4

Months 7–9 · Consolidate to $25k

Target $10,000–12,000 · all self-funded

17 **Negotiate tipping fees from 2–3 waste suppliers**

Once you have 3–6 months of operating history and demonstrated waste processing capability, approach food businesses with a formal proposal: pay you a tipping fee of $30–80/tonne to collect their organic waste. This is well below commercial waste removal costs. Even one food processor paying $50/tonne for 2 tonnes/week adds $400–500/month in pure additional income. Frame it as a waste management service contract, not a favour.

$400–500/month additional **Month 6–7**

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18 **Add subscription and standing order incentives**

Offer a 10% discount for monthly standing orders paid upfront via PayID. One customer committing to $80/month is operationally worth ten casual $20 buyers. Frame it as a 'farm subscription box': 1 kg live larvae + 500g frass, delivered or local pickup monthly. This smooths your production planning, reduces spoilage, and locks in revenue before each harvest cycle.

10% standing order discount **Month 5 onwards**

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19 **Begin documenting the protocol for future licensing**

By month 7–8 you will have a proven, reproducible system with real yield data, customer testimonials, and regulatory confirmation. Start compiling a formal operations manual: substrate sources and approval records, rearing conditions, yield data by batch, processing steps, packaging specifications, customer agreements. This document is the asset you eventually licence (at $20,000–50,000 per licensee) to operators in non-competing geographies. Your microscope photographs and batch records are already building this without extra effort.

No cost **Month 7–9** **Licensing value: $20k–50k/licensee**

Revenue Model — Path to $25,000 by December 2026

REVENUE STREAM	UNIT PRICE	VOLUME TARGET	PERIOD	PROJECTED
Live BSFL — direct & local pet stores	$24/kg wholesale	3→8 kg/wk	May–Dec	**~$6,000**
Dried BSFL — online & markets	$70/kg	1→3 kg/wk	Jul–Dec	**~$5,500**
Frass fertiliser — retail & farm	$12/kg retail / $9/kg bulk	10→30 kg/wk	Jun–Dec	**~$5,000**
Farmers market stalls (fortnightly)	$500/day avg	2× per month	Jun–Dec	**~$5,500**
Tipping fees from waste suppliers	$50/tonne	1–2 t/wk from month 6	Oct–Dec	**~$2,000**
Online wholesale retailer (1 account)	$20/kg	3 kg/wk from month 5	Sep–Dec	**~$2,500**
Chitin powder — agricultural grade	$25/kg avg	0.5→1.5 kg/wk from month 5	Sep–Dec	**~$1,200**
TOTAL PROJECTED				**~$27,700**

Startup Budget — $500 Ceiling

ITEM	COST	NOTES
ABN / Company registration	$0–$576	Free for partnership ABN; $576 ASIC fee for Pty Ltd (recommended)
Business name (ASIC)	$42	1-year registration; renew annually
Solicitor — partnership or shareholders agreement	$500–800*	*Funded separately from startup budget. Most important spend in the business.
Website — GitHub Pages + domain	$15/year	GitHub Pages is free forever. Domain (~$15/yr) via Namecheap or Cloudflare. No Shopify fees.
Packaging — kraft bags × 200, labels	$80	500g and 1kg zip-lock kraft bags; labels printed at home via Canva (free)
USB microscope (50–200x)	$30	Highest ROI purchase. Content creation + quality control + expertise signalling.
Used food dehydrator	$55	Facebook Marketplace or Gumtree. Buy now — dried BSFL available from week 1.

Product liability insurance (1st year)	$180	Required before first product sale. Rural/agricultural insurers: Elders, Gallagher.
Printing — product sheets, market signage	$20	A3 microscope photo for market stall backdrop; one-page wholesale sheet
First market stall fee	$50	Lockyer Valley or Toowoomba farmers market day fee
Contingency	$28	
TOTAL	**$500**	Solicitor cost funded separately from early revenue

High-Impact Low-Cost Additions

ADDITION	COST	IMPACT	HOW TO ACTION
USB microscope + content strategy	~$30	Expertise positioning, premium pricing justification, viral content in reptile communities, quality control tool	Buy on Amazon. Photograph larvae weekly. Post in Facebook groups and on your website. Print best image as A3 market stall signage.
Batch tracking spreadsheet (public)	$0	Traceability builds trust with reptile keepers and organic farmers. Simultaneously builds licensing protocol documentation.	Track: substrate source, harvest date, temp range, yield weight per batch. Pin a monthly summary to your Facebook page.
Lockyer Valley Growers Association outreach	$0	Puts you in front of 20–50 frass buyers at once. Contract sizes 10–20x larger than retail. One contract = $5,000.	Attend one grower network meeting. Pitch as local food waste processor producing organic soil amendment. Bring sample bags.
Pre-sell before first harvest	$0	Discovers real demand before you invest time. Deposits fund early costs. Pulls first revenue forward by 4–6 weeks.	Post in Facebook groups now: 'Lockyer Valley BSFL launching in 6 weeks — early access list?' Collect names and PayID deposits.
Standing order subscriptions (10% discount)	$0	Smooths production planning, reduces spoilage, locks in revenue before each cycle. One standing order = ten casual sales in certainty.	Offer from first sale. 'Monthly farm subscription box: 1 kg live + 500g frass, 10% off, pay upfront by PayID.'

GST registration from day one	$0	Lets you claim back GST on all equipment. Required by farm and business customers who want tax invoices.	Register simultaneously with ABN via Business Registration Service. Use Wave Accounting (free) to track.

Chitin & Chitosan — The Hidden Revenue Stream

Chitin is the second most abundant biopolymer on earth after cellulose. It is already inside every batch you produce — in the larval exoskeletons shed during moulting, in pupal casings, and in the frass. Almost every BSFL operator globally discards it. Extracting and selling it requires no additional biological work and adds a fourth revenue stream with a higher margin per kilogram than any other product in the business.

> The key insight: you do not need to sacrifice larvae to produce chitin. Your frass — already being produced as a byproduct of every rearing cycle — contains approximately 5–8% chitin by dried weight, derived from shed exoskeleton material and pupal casings. A standard 50 kg/week rearing operation producing ~25 kg/week of dried frass yields approximately 1.5–2 kg of crude chitin per week with no change to your rearing process whatsoever. BSF-derived chitosan is described by international specialty suppliers as the gold standard in chitosan quality, commanding a significant premium over crustacean-derived equivalents due to its higher deacetylation degree, consistent substrate, and clean insect origin.

Chitin Yield and Revenue — By Operation Scale

OPERATION SCALE	FRASS PRODUCED/WK	CRUDE CHITIN YIELD/WK	ANNUAL CHITIN (kg)	AGRIC. GRADE $25/kg	SPECIALTY GRADE $75/kg	PREMIUM BSF CHITOSAN $120/kg
Year 1 (Site 1 pilot)	~25 kg/wk	~1.5–2 kg/wk	~80–100 kg/yr	**~$2,200/yr**	**~$6,750/yr**	**~$10,800/yr**
Site 1 at full scale (~500 t/yr waste)	~200 kg/wk	~12–16 kg/wk	~700 kg/yr	**~$17,500/yr**	**~$52,500/yr**	**~$84,000/yr**
5-site network (~13,000 t/yr waste)	~5,000 kg/wk	~300–400 kg/wk	~18,000 kg/yr	**~$450,000/yr**	**~$1.35M/yr**	**~$2.16M/yr**

Chitin yield assumes 6% chitin content of dried frass. Agricultural grade = crude chitin powder sold to vegetable growers. Specialty grade = purified chitin suitable for horticultural biostimulants. Premium = fully processed BSF-derived chitosan for nutraceutical or industrial buyers. Revenue ceiling requires quality documentation and buyer relationships that build over 12–18 months.

The Extraction Process — Simple Enough for Month 4

The standard chemical chitin extraction is a three-step bench process. All reagents are standard agricultural/industrial chemicals. At small scale, no purpose-built equipment is required beyond what most farms already have or can obtain cheaply.

STEP	PROCESS	REAGENT	CONDITIONS	PURPOSE
1 — Demineralisation	Dissolve calcium carbonate from frass into acid solution	5% hydrochloric acid (HCl)	Room temp, 2–4 hrs, stir occasionally	Removes mineral content. Frass fizzes as calcium dissolves.

2 — Deproteinisation	Dissolve protein content from remaining solids	5–10% sodium hydroxide (NaOH)	60–80°C, 2–4 hrs under stirring	Removes protein fraction. Produces crude chitin residue after washing and drying.
3 — Deacetylation (optional for chitosan)	Convert chitin to more soluble chitosan form	50% NaOH (concentrated)	90–100°C, 4 hrs. Wash to neutral pH after.	Produces chitosan: more soluble, higher bioactivity, higher price. Skip for agricultural-grade chitin powder.
4 — Wash, dry, mill	Rinse to neutral pH, dry, grind to powder	Water, food dehydrator (already owned)	Wash 6–8 times. Dry at 60°C overnight.	Final product: light-coloured powder. Package in sealed kraft bags with batch number and substrate declaration.

Setup Cost — Month 4 Entry

ITEM	COST	NOTES
Food-grade hydrochloric acid (5L)	~$25–40	Agricultural supplier or pool acid (33% HCl). Dilute to 5% working solution.
Sodium hydroxide — NaOH pellets (2 kg)	~$15–25	Pool/spa chemical supplier or agricultural outlet. Same chemical used for biodiesel transesterification later.
Stainless steel pot (10–20L)	~$30–50	Second-hand from op shop or kitchen supplier. Must be stainless — acid will corrode aluminium.
Food-grade filter cloth / muslin	~$10	For filtering the chitin residue from solution. Multiple uses per metre.
pH paper strips (wide range)	~$10	To confirm neutral pH after washing. Essential — acidic residue damages product quality and buyer trust.
Nitrile gloves, safety goggles, apron	~$20–30	Required. HCl and concentrated NaOH are corrosive. Standard farm chemical handling precautions apply.
Kraft bags + labels (chitin specific)	~$25	Same supplier as larvae bags. Label: batch number, frass substrate source, extraction date, grade.
TOTAL SETUP	**~$135–$190**	Reagents replenished from revenue. Food dehydrator (already owned) used for drying.

The Market Pathway — Three Stages of Value Escalation

STAGE	PRODUCT	TARGET BUYER	PRICE/KG	WHEN ACHIEVABLE	WHAT ENABLES IT
1 — Agricultural chitin powder	Crude dried chitin from BSFL frass, lightly purified	Lockyer Valley vegetable growers, organic farmers, market gardeners already buying your frass	**$15–40/kg**	Month 4–5 Year 1	Basic extraction process, batch documentation, existing customer relationship
2 — Specialty horticultural chitosan	Purified chitosan with documented deacetylation degree and substrate traceability	Premium organic farms, hydroponic operations, nursery chains, agricultural distributors	**$50–80/kg**	Month 12–18 Year 2	Consistent batch quality, independent lab test certificate, APVMA product notification
3 — Premium BSF chitosan (nutraceutical / industrial)	High-purity insect-derived chitosan; substrate-traceable BSF-origin certificate	Nutraceutical formulators, cosmetics manufacturers, industrial water treatment, specialty chemical distributors	**$100–150/kg**	Year 2–3 Site 1 at scale	Documented substrate consistency, high deacetylation verification (>75% DD), accredited lab analysis, food-grade facility certification for handling

The strategic advantage of starting with agricultural-grade chitin sold to your existing frass customers is that it builds the trust and documentation trail needed for the higher-grade markets, without any additional marketing effort or customer acquisition cost. Your frass buyers already understand the product, already trust the substrate sourcing, and are already looking for ways to reduce synthetic inputs. The pitch for Stage 1 is simply: 'This is the active fraction of the frass, isolated. Same substrate, higher concentration, better penetration into soil.' By the time you're ready to approach specialty and nutraceutical buyers at Stage 3, you have 12–18 months of batch records, an independent lab certificate, and a customer base who can provide references. BSF-derived chitosan commands this premium specifically because of traceable, controlled substrate — and you are building that documentation from day one.

Regulatory Documents Checklist — Queensland

All items below are specific to Queensland and the Lockyer Valley. Items marked Required must be completed before the associated activity begins.

DOCUMENT	COST	WHEN	AUTHORITY
Tax File Number (TFN)	Free	Before anything else	ato.gov.au
ABN — company or partnership	$0–$576	Day 1 — required	abr.gov.au / asic.gov.au
Business name registration	$42/yr	Week 1 — if using trade name	asic.gov.au
Shareholders / partnership agreement	$500–800	Before operating — critical	Solicitor
GST registration (ATO)	Free	Day 1 recommended	ato.gov.au
LVRC Environmental Health Officer consultation	Free	Week 1–2 — before construction	lockyervalley.qld.gov.au 1300 005 872
Biosecurity QLD — substrate approval confirmation	Free	Week 2 — before waste contracts	daf.qld.gov.au 13 25 23
Waste supply contracts (with waste generators)	$0–$300	Before accepting commercial waste	Self-drafted or solicitor
Product liability insurance	~$180–300 /yr	Before first product sale	Elders, Gallagher, Primacy
Animal feed notification — DAF Queensland	TBC	Before poultry/aquaculture feed sales	daf.qld.gov.au
Product labels — APVMA compliant	~$500–1,000	Before retail/wholesale sale	Canva for design; solicitor review

Partner Roles & Equity Structure

This partnership pairs two complementary and unusually well-matched skill sets. The division below is designed so that all physical work and the majority of total work sits with the on-farm partner, while the Brisbane-based partner contributes high-leverage systems and strategic work remotely.

> Equity split: 51% (you) / 49% (partner). The 51/49 structure gives you operational control — any deadlock defaults to your decision — while keeping the split close enough to signal genuine equal partnership to customers, regulators, and future investors. It is deliberately not 60/40 or 65/35: a wider split would undervalue your partner's founder experience and IT systems capability, which are worth more to this business than hours alone. The 2% margin is a tie-breaker, not a statement of relative worth. This split should be reviewed at the 12-month mark based on actual contribution.

YOUR ROLE

Managing Director / Chief Operating Officer

51% equity · All physical work · Majority of total work

Strengths you bring: Double degree in management and biological sciences — almost precisely the qualification you would design for a BSFL operation. Your biological sciences background means you can read peer-reviewed literature directly, optimise rearing conditions from first principles, and speak credibly to researchers and regulators. Your management training means you run the commercial operation without outsourcing strategic decisions. Most small insect farms are run by enthusiastic hobbyists with no business training, or business people with no biology. You have both, which is a genuine competitive moat.

Operational responsibilities:

- All rearing, harvesting, quality control, and substrate management
- Waste supplier relationships and collection logistics
- Farmers market presence and local wholesale account management
- Batch tracking, protocol documentation, and substrate optimisation
- Scientific record-keeping — the core licensing asset
- Local regulatory compliance (LVRC, DAF, biosecurity)
- Product development and frass processing
- Financial oversight via Wave Accounting (~30 min/week)
- Local networking — Lockyer Valley Growers Association
- All strategic decisions: you hold casting vote at 51%

Strategic responsibilities:

- Overall business strategy and direction
- Licensing model development (months 18–36)
- Investor or lender relationships if sought in future

PARTNER ROLE

Chief Technology Officer / Growth Systems

49% equity · Remote from Brisbane · ~10–15 hrs/week initially

Strengths he brings: His real value is not generic IT support — it is two specific things. First, he knows how to build systems that scale without the founder becoming the bottleneck. Second, he has been through company formation before, which means he has already made (or nearly made) the expensive mistakes around equity, governance, and hiring. That experience is worth more than any technical skill, and it should be used at the moments that matter most: structuring the shareholders agreement and designing the eventual licensing model.

Technology responsibilities:

- GitHub Pages website build and maintenance
- E-commerce flow: PayID integration, order management, automated emails
- CRM setup — Notion database tracking every customer, order, and standing agreement
- Bookkeeping setup in Wave Accounting; BAS preparation support
- Company secretarial obligations: ASIC annual reviews, minutes, share register
- Any paid advertising or SEO when revenue justifies it
- Technology platform selection and onboarding
- Online wholesale retailer account applications and onboarding
- Social media scheduling and content calendar (using your microscope photos)

Founder experience applied to:

- Shareholders agreement review — he knows which clauses matter from experience
- Licensing and franchise agreement structure (months 18–36)
- Investor or lender pitch materials if sought

Scope boundary: Your partner's remit explicitly excludes customer relationships, supplier negotiations, and anything requiring physical presence on the farm. Those stay with you — because you are there, because local trust is not transferable remotely, and because your qualifications give you the credibility to own those relationships fully. His systems multiply your physical output; they do not substitute for it.

Critical IP clause for the shareholders agreement: The rearing protocol, batch documentation, and biological methods developed on the farm must be owned by the company — not by either individual. Your sweat equity contribution to developing that IP must be explicitly recognised in the 51% equity split. Your partner's founder experience means he will understand why this clause matters; make sure you both read it the same way before signing anything.

Key Risks & Mitigations

RISK	LIKELIHOOD	MITIGATION
Summer overheating (Oct–Mar) wipes a production cycle	High in Lockyer Valley	$20 thermometer + shade cloth before November. BSFL mortality accelerates above 38°C. Plan rearing shed ventilation now.
Partner dispute without formal agreement	Critical risk	Shareholders agreement or partnership deed before operating. Cover: IP ownership, profit split, exit provisions, decision-making.
Slow customer acquisition — no sales by week 8	Manageable	Pre-sell via Facebook groups before first harvest. Lock in at least one wholesale account before launch. Market stall generates immediate cash.
Substrate contamination invalidates feed approval	Low if managed	Get written DAF substrate approval before signing waste contracts. Batch tracking spreadsheet documents every input. Reject any waste with unidentified animal material.
Live larvae die in transit — refund liability	Moderate	Prioritise dried products for non-local sales. Use Express Post for live orders. Clear refund policy on website. This risk disappears entirely for local pickup sales.

Long-Term Vision — Scale-Up to 30% of SEQ Food Waste

The $25,000 proof-of-concept milestone is the foundation. The long-term opportunity is defined by one number: how much organic food waste South East Queensland generates, and what capturing a meaningful share of it is worth.

The SEQ Food Waste Opportunity

> SEQ has a population of approximately 4.2 million people (2025), growing at ~3% per year toward a projected 5.9 million by 2050. Queensland as a whole generated 10.25 million tonnes of total waste in 2024-25, of which SEQ accounts for ~73% by population. Australia generates 7.6 million tonnes of food waste annually at a national level — roughly 300 kg per person per year — with approximately two-thirds coming from commercial and consumer-facing businesses. Applying Queensland's share (~13% of national population) and SEQ's share (~73% of Queensland), the SEQ organic food waste pool is estimated at approximately 720,000–850,000 tonnes per year. Less than 30% of household waste is currently diverted from landfill in Queensland, well below the state target of 55% by 2025 — meaning the majority of this stream remains available and landfill operators are actively seeking alternatives.

METRIC	FIGURE	SOURCE
SEQ population (2025 est.)	**~4.2 million**	ABS / QGSO 2023 ERP + 3% growth rate
SEQ projected population (2050)	**5.9 million**	McCrindle / Queensland Gov projections
Australia annual food waste	**7.6 million tonnes/yr**	National Food Waste Strategy 2021; Foodbank 2024
Food waste per capita (Australia)	**~300 kg/person/yr**	DCCEEW / Fight Food Waste CRC
Queensland total waste generated (2024-25)	**10.25 million tonnes/yr**	Qld Recycling and Waste Report 2024-25
SEQ share of Queensland population	**~73%**	QGSO 2023: 3.98m of 5.45m Queenslanders in SEQ
SEQ organic food waste pool (est.)	**~720,000–850,000 t/yr**	Derived: national per-capita × SEQ population
Household waste diverted from landfill — QLD	**28.3% (target: 55%)**	Qld State of the Environment Report 2024
Commercial/industrial waste diverted — QLD	**54.6% (target: 65%)**	Qld State of the Environment Report 2024

Scale-Up Model: 1% to 30% of SEQ Food Waste

The table below models what capturing increasing shares of SEQ's ~785,000 tonne/year food waste pool produces in output and revenue, using conservative bioconversion yields: 20% larval biomass from substrate, 30% dried meal from fresh larvae, 50% frass from substrate weight, and pricing at wholesale rates. The 30% capture scenario is the realistic ceiling for a company-owned network of BSFL sites across SEQ — a coordinated multi-location operation, all under one Pty Ltd.

CAPTURE %	WASTE PROCESSED (t/yr)	FRESH LARVAE BIOMASS (t/yr)	DRIED MEAL (t/yr)	FRASS (t/yr)	TIPPING FEE INCOME	PRODUCT REVENUE (est.)	TOTAL REVENUE (est.)
0.001% (Year 1 — single farm)	~8 t/yr	1.6 t/yr	480 kg/yr	4 t/yr	~$0	~$25,000	**~$25,000**
1%	7,850 t/yr	1,570 t/yr	471 t/yr	3,925 t/yr	$2.4M/yr ($50/t × 47k t)	Meal: $1.7B Frass: $39M	**~$43M/yr**
5%	39,250 t/yr	7,850 t/yr	2,355 t/yr	19,625 t/yr	$11.8M/yr	Meal ~$8.7B Frass ~$196M	**~$215M/yr (network)**
10%	78,500 t/yr	15,700 t/yr	4,710 t/yr	39,250 t/yr	$23.6M/yr	Meal ~$17.4B Frass ~$393M	**~$430M/yr (network)**
30% (network ceiling)	235,500 t/yr	47,100 t/yr	14,130 t/yr	117,750 t/yr	$70.7M/yr	Meal ~$52B Frass ~$1.2B	**~$1.3B/yr (owned network)**

Revenue figures use wholesale rates: dried BSFL meal at $3,700/t (World Aquaculture, 2025), frass at $10/t bulk wholesale. Tipping fee at $50/t received. Meal revenue at network scale reflects the full theoretical value of the output — realistically, only a fraction of this would be captured at wholesale vs retail. The intent is to show the ceiling of the resource, not a sales projection.

The Four Stages from Farm to Network

STAGE	TIMELINE	DESCRIPTION	KEY MILESTONE	REVENUE RANGE
1 — Proof of concept (single farm)	Apr–Dec 2026	One farm, live and dried BSFL, frass, direct-to-consumer and local wholesale. Lockyer Valley only. All physical work on-farm.	$25,000 revenue. Documented batch records. First wholesale account.	**$25,000**
2 — Regional expansion (direct + licensing)	2027–2028	Scale own farm to 50–100 kg larvae/week. First 2–3 licensed operators in non-competing SEQ postcodes. Frass contracts with Lockyer Valley growers. Online wholesale account live.	$200k–400k/yr own farm. First licensee fee ($20–50k each). Royalties begin.	**$250k–$500k/yr**

3 — SEQ licensed network (5–10 operators)	2028–2031	5–10 licensed operators across SEQ processing combined 1,000–5,000 t/yr of food waste. Centralised protocol, IP, and quality standards owned by the company. Royalties on all output.	Protocol recognised as the SEQ standard for BSFL bioconversion. Frass supply agreements with vegetable growers at scale.	**$1M–$5M/yr (own + royalties)**
4 — Biodiesel / SAF layer (lipid extraction)	2030+	Once network volume reaches 5,000+ t/yr substrate, lipid extraction becomes viable. Centrally process larval oil from multiple operators into biodiesel or SAF precursor. Feedstock cost is effectively zero — larvae funded by protein meal sales.	First commercial biodiesel batch sold to fleet or marina customer. Carbon credit registration for the network.	**Additional $500k–$2M/yr (fuel + carbon credits)**

Owned Multi-Site Expansion — The Company Operates All Locations

Rather than licensing the protocol to third parties, the company builds, owns, and operates every facility. This preserves complete control over product quality, brand, waste supply relationships, and revenue — at the cost of significantly higher capital requirements as each site is added. The constraint shifts from intellectual property to capital and management bandwidth. The model that makes this work is the same as all capital-intensive infrastructure businesses: use operating cashflow and grant funding from early sites to fund subsequent sites, and use the Pty Ltd structure to raise debt against proven assets.

> Industry CAPEX benchmarks: The most efficient large-scale BSF facility globally (Entobel, Vietnam) runs at ~USD $2,700 per tonne of annual dried insect production capacity — achieved through low-cost construction and labour in Southeast Asia. Protix's flagship Netherlands facility runs at ~USD $19,000 per tonne capacity. A realistic Australian mid-range for a purpose-built but non-automated regional facility is AUD $6,000–$10,000 per tonne of annual dried larvae capacity. A modular approach using converted industrial sheds in semi-urban SEQ (Ipswich, Logan, Sunshine Coast, Gold Coast fringe) keeps build cost at the lower end of this range. For comparison, Goterra's MIBS shipping-container units — the closest Australian commercial analogue — process ~10 tonnes of food waste per month per unit. Industry insight: most large insect farm failures resulted from building industrial scale before validating unit economics. The owned multi-site model below deliberately delays capital deployment until each prior site is profitable.

Standard Site Specification — Per Location

Each owned site is designed around a standard repeatable module: a 1,000–2,000 m² leased or purchased industrial shed, climate-controlled rearing rooms, central processing (drying, milling, packaging), frass handling, and a dedicated waste receiving bay. The Lockyer Valley anchor farm is Site 1. Subsequent sites target industrial estates adjacent to major food waste generators within 30 km of SEQ population centres.

ELEMENT	SPECIFICATION	COST ESTIMATE (AUD)	NOTES
Site lease — industrial shed	1,000–2,000 m² in semi-urban SEQ industrial estate	$60,000–$120,000/yr ($5–8/m²/month)	Preferably adjacent to food manufacturer, supermarket DC, or council transfer station
Rearing infrastructure build-out	Climate-controlled rooms, shelving, IBC containers, ventilation, thermostat control	$80,000–$150,000	Replicates Lockyer Valley proven setup at scale. Modular — add capacity incrementally.
Drying and processing equipment	Industrial dehydrator/drum dryer, grinder/mill, frass separator, packaging line	$60,000–$120,000	Second-hand industrial food processing equipment dramatically reduces this cost. Budget for new at commercial scale.
Waste receiving and pre-processing	Receiving bay, grinder/shredder for waste prep, storage bins	$30,000–$60,000	Reduces labour time significantly. Essential at >50 t/week intake volumes.

Colony breeding room	Dedicated adult fly breeding space; seeded from Lockyer Valley parent colony	$15,000–$30,000	Site 1 Lockyer Valley becomes the central breeding hub supplying eggs to all sites.
IoT monitoring and control	Temperature, humidity, CO_2 sensors; remote monitoring dashboard (your partner builds)	$5,000–$15,000 per site	Remote monitoring means your partner in Brisbane manages all sites without being on-farm.
Permits, compliance, insurance setup	EPA/DES notification, council approvals, product liability insurance, food safety program	$5,000–$15,000	Decreasing marginal cost per site as approvals are template-based from Site 1.
Working capital (3 months operations)	Labour, utilities, packaging, waste transport for first quarter before revenue stabilises	$40,000–$80,000	Funded from prior site cashflow or debt against Site 1 assets.
TOTAL CAPEX PER SITE (est.)		**$235,000–$590,000**	Use $300–400k as planning figure for a 50 t/week processing site.

Owned-Site Rollout Plan — 2026 to 2032

Each site is funded by a combination of retained earnings from earlier sites, government grants (SEQ City Deal infrastructure funding, ARENA low-carbon fuels grants), and when required, commercial debt secured against proven operating assets. No new equity is raised — ownership remains 51/49 throughout. The Lockyer Valley site is the proof-of-concept and the permanent breeding hub. All subsequent sites are rearing-and-processing only, supplied with eggs from Site 1.

SITE	LOCATION (SUGGESTED)	TIMING	WASTE INTAKE	CAPEX (PLAN)	FUNDING MECHANISM	ANNUAL REVENUE (est.)	STAFF (SITE)
Site 1 Lockyer Valley (anchor + breeding hub)	Gatton / Laidley own farm	Apr 2026 (active)	Pilot scale ~8–50 t/yr	$500 (proof of concept)	Personal capital, tipping fees, product revenue	**$25k–$200k (Year 1–2)**	You + part-time
Site 2 Ipswich corridor	Ipswich industrial estate or Riverview	2027	~500–1,000 t/yr	~$300k	Site 1 cashflow + SEQ City Deal grant application	**~$400k–$700k/yr**	2–3 FTE
Site 3 Logan / Beaudesert	Carole Park or Loganholme industrial	2028	~1,000–2,000 t/yr	~$350k	Sites 1+2 cashflow + ARENA bioenergy grant	**~$700k–$1.2 M/yr**	3–4 FTE

Site 4 Sunshine Coast fringe	Warana / Kunda Park industrial	2029	~2,000–3,000 t/yr	~$400k	Retained earnings; debt against Sites 1–3 assets	**~$1.2M–$1.8M/yr**	4–5 FTE
Site 5 Gold Coast fringe	Arundel / Molendinar industrial	2030	~3,000–5,000 t/yr	~$500k	Retained earnings; council waste processing contract	**~$1.8M–$2.8M/yr**	5–6 FTE
Central processing hub (biodiesel + SAF)	Lockyer Valley or Ipswich — adjacent to Site 1 or 2	2031–2032	Lipid from all 5 sites ~500–1,000 t/yr	~$800k–$1.5M (grant-funded)	ARENA + Qld Govt biofuels mandate grant	**+$1M–$3M/yr (fuel + carbon credits)**	2–3 FTE (processing)
TOTAL AT MATURITY (Sites 1–5 + hub, ~2032)	~10,500–13,000 t/yr food waste processed	~1.4% of SEQ food waste stream		**~$2.4M total CAPEX deployed**		**~$8M–$12M/yr combined revenue**	~20–25 FTE total

Revenue per site estimated at: live/dried BSFL $24–70/kg wholesale, frass $9–12/kg, tipping fees $50/t waste received. Each site's revenue depends on its waste intake, local market mix (wholesale vs retail vs farm contract), and product split. The maturity figure of $8–12M/yr assumes a blended average of $800–1,000/tonne of food waste processed across all revenue streams.

Management Structure at Scale

Running five owned sites requires a management structure that preserves the founding partners' control while delegating day-to-day site operations. The model below keeps the two founders in senior operational and strategic roles, with site managers hired progressively from retained earnings.

ROLE	PERSON	RESPONSIBILITIES AT SCALE	WHEN HIRED/ACTIVE
Managing Director / COO	**You (51%)**	Overall operations, biological quality standards, waste supply relationships, Site 1 direct management, new site build-out oversight, regulatory compliance across all sites.	Day 1 — permanent
CTO / Systems & Growth	**Partner (49%)**	Remote IoT monitoring of all sites, financial oversight, grant applications, council contracts, new site feasibility, product sales platforms, company secretarial.	Day 1 — permanent
Site Manager (per site from Site 2)	Hired FTE	Day-to-day rearing, harvest, waste receiving, local customer relationships, product despatch. Reports to you as COO.	Hired when site opens. ~$65–80k/yr salary.

Processing / Quality Officer	Hired FTE (from Site 3)	Cross-site quality control, batch testing, frass and dried larvae certification, food safety program management.	Month of Site 3 opening. ~ $70–85k/yr.
Sales & Customer Manager	Hired FTE (from Site 3–4)	Wholesale account management, market stall coordination, online retail, new customer acquisition. Reports to partner.	When revenue justifies. ~$65–75k/yr.

Capital Strategy — No External Equity Required

The critical discipline in the owned multi-site model is sequencing capital deployment. The failure mode for the global insect farming industry has consistently been building industrial-scale facilities before unit economics were validated at smaller scale. AgriProtein, Ynsect, and Beta Hatch all raised $10M–$100M+ and deployed it into large infrastructure before proving that the operating economics worked — then failed. Your sequencing inverts this: prove the model on $500, scale Site 1 to $200k/yr revenue, use that cashflow and a single government grant application to fund Site 2, and so on. No external equity means 51/49 ownership remains intact at every stage. The SEQ City Deal ($54M organics infrastructure), ARENA ($1.7B low-carbon fuels), and the Recycling Modernisation Fund are all grant programs that a demonstrated operating processor — not a startup — can access. This is why Site 1 must generate documented yield data, council relationships, and operating history before Site 2 capital is deployed.

Revised Scale Model — Owned Sites at 30% SEQ Capture

The table below recalculates what 30% SEQ food waste capture looks like as a fully-owned company rather than a licensed network. At 30% capture (~235,000 t/yr), you would need approximately 45–60 owned sites of the standard specification above. This is a 20–30 year build, not a 10 year plan. The more realistic and compelling case is the 5-site, 1.4% capture scenario above — which produces $8–12M/yr with full ownership, no licensing complexity, and direct control of every product that leaves every site.

SCENARIO	WASTE PROCESSED	% OF SEQ FOOD WASTE	SITES REQUIRED	TOTAL CAPEX	TIMELINE	ANNUAL REVENUE
Proof of concept (Site 1)	~8–50 t/yr	0.001%	1	$500	2026	**$25k–$200k**
Early commercial (Sites 1–2)	~1,500 t/yr	~0.2%	2	~$300k total	2027	**$600k–$900k**
Regional operator (Sites 1–5)	~10,500–13,000 t/yr	~1.4%	5 + hub	~$2.4M total	2030–2032	**$8M–$12M/yr**
Major SEQ operator (Sites 1–15)	~40,000–50,000 t/yr	~5–6%	15 + hub	~$6–8M total	2033–2038	**$28M–$40M/yr**

Dominant SEQ processor (30% capture)	~235,000 t/yr	30%	45–60 sites + 3 hubs	~$25–40M total	2040s+ (generational)	**$160M–$250M/yr**

> Practical focus: The 5-site, $8–12M/yr scenario is the realistic 10-year ambition for two founders who want to own and operate everything themselves. It requires approximately $2.4M in cumulative capital deployed over 6 years — entirely self-funded from cashflow and grants if execution is disciplined. Beyond 15 sites, the management complexity of operating diverse sites across SEQ argues either for external management hires (increasing OPEX significantly) or for a partial pivot to a hybrid own-and-license model for distant sites, with the core Lockyer Valley / Ipswich / Logan cluster remaining fully owned. That decision belongs to 2033, not 2026.

Policy and Funding Context

POLICY / FUNDING	RELEVANCE TO THIS BUSINESS
QLD Waste Reduction and Recycling Act 2011 — 55% household diversion target	Currently at 28.3%. Gap = 26.7 percentage points. Every tonne of food waste you divert counts toward council targets. Councils have financial incentives to direct waste streams to compliant processors. Council waste processing contracts are available to operated (not licensed) facilities.
SEQ City Deal — $54M organics infrastructure allocation	Explicitly allocated to organics processing infrastructure across SEQ. A fully-owned, operating BSFL facility with documented yield data and council relationships is a stronger grant applicant than an unlaunched startup.
Recycling Modernisation Fund — Australian Government	Supporting 25 projects diverting 1.1M tonnes from landfill. BSFL bioconversion qualifies as resource recovery infrastructure. Applications are assessed on demonstrated operating capability — Site 1 builds that record.
Australia Circular Economy Framework (2024)	Federal blueprint prioritising food and agriculture waste streams. Multi-site owned processors are exactly the infrastructure scale the framework is designed to support.
ARENA — $1.7B allocated to low-carbon liquid fuels (2024 Federal Budget)	The central biodiesel/SAF hub at Sites 5+ is an ARENA-eligible project. A demonstrated waste processing network with yield data, regulatory approvals, and offtake agreements is required before application.
QLD waste levy — $75/tonne	Landfill disposal costs businesses $75+/t in Queensland. A tipping fee of $50/t positions you as cheaper than landfill. At 5 sites processing 13,000 t/yr, you are saving your waste suppliers $975,000/yr in avoided levies — a powerful commercial proposition.

> The $25,000 target is not the goal — it is the evidence. Evidence that the rearing protocol works at commercial volume, that customers pay without prompting, that the frass story lands with local farmers, and that two partners can execute a complex biological and commercial operation reliably. That evidence is what makes Site 2 fundable, what makes a grant application credible, and what makes a council waste processing contract winnable. Start with what you know, document everything, operate with discipline, and let the scale follow at a pace the cashflow supports.

This plan was prepared using current market data (April 2026) and peer-reviewed scientific literature on Hermetia illucens commercial production. SEQ food waste figures are derived from Queensland Government waste reporting (2024-25), ABS population data, and national food waste estimates from the National Food Waste Strategy Feasibility Study (2021) and Foodbank Australia (2024). Revenue projections are estimates based on comparable Australian BSFL operations and current wholesale/retail pricing. All regulatory information is current as of April 2026 and should be independently verified before operating.